

Chapter 10: Compare and Contrast Security Implications of Different Architecture Models

Securing the Network

- A multi-layered defense using **firewalls**, **ACLs (Access Control Lists)**, **IDS (Intrusion Detection Systems)/IPS (Intrusion Prevention Systems)**, and **SIEM (Security Information and Event Management)** is essential for robust protection.
- **Firewalls** with default "deny all" rules and specific **allow lists** ensure traffic control.
- **ACLs** filter traffic based on defined criteria; **IDS/IPS** detect and prevent malicious activity in real-time.
- **SIEM** aggregates logs and alerts from across the network, offering a unified view and faster threat response.

Securing the Servers

- Key servers like domain controllers and SQL servers must be protected as they handle authentication and sensitive data. Mail servers and video conferencing platforms are common attack targets. Cloud storage solutions (e.g., AWS S3, Azure, Google Cloud) offer built-in encryption, access control, and monitoring to enhance security.

Securing the Hosts

- Host security involves protecting endpoints such as user devices with antivirus software, endpoint detection and response (EDR) tools, and mobile device management (MDM) systems. Multifactor authentication (MFA) strengthens access control by requiring multiple forms of identity verification, reducing the risk of unauthorized access.

Infrastructure as Code (IaC)

- IaC uses scripts (e.g., YAML, JSON) to automate and manage infrastructure, replacing manual processes.
- Enhances **efficiency**, **consistency**, and **scalability** across environments.
- Supports **version control**, team collaboration, and infrastructure reproducibility.
- Tools: **Terraform**, **Ansible**, **Puppet**, **Chef**; supported by AWS, Azure, and Google Cloud. These tools help organizations build scalable, reliable, and efficient IT systems.

Serverless

- Eliminates server management—CSPs (Cloud Service Providers) handle infrastructure, scalability, and security.
- Developers focus on deploying code, while **BaaS (Backend as a Service)** handles backend functions like authentication and storage.
- Offers **cost efficiency** (no hardware investment), but the customer remains responsible for application-level security and data protection.

Internet of Things (IoT)

- IoT includes smart devices with sensors and connectivity, enabling real-time monitoring and automation.
- Security concerns:
 - **Lack of standardization, insecure communication, and data privacy risks.**
 - Vulnerabilities due to **long device lifecycles, physical access, and global supply chains.**
 - **User awareness** and manufacturer responsibility are critical for proper device configuration and maintenance.

SCADA (Supervisory Control and Data Acquisition)

- A type of **Industrial Control System (ICS)** used in industries like energy, manufacturing, and logistics.
- **Figure 10.6** illustrates a 4-level SCADA architecture:
 - **Level 0 (Plant Field Level):** This is the lowest level in the SCADA system hierarchy. It includes the physical equipment and processes on the factory floor, such as sensors, actuators, motors, pumps, and other industrial devices. These devices gather data and perform actions as directed by the higher level controllers.
 - **Level 1 (Controller Level):** This level is responsible for the real-time control of the physical processes. It includes devices such as **Programmable Logic Controllers (PLCs)** that receive input from sensors on the plant floor, process the data, and send commands to actuators and other devices to control the industrial processes. Level 1 controllers ensure that the plant operates efficiently and safely.
 - **Level 2 (Coordinating Computer Level):** At this level, there are **supervisory computers or Human-Machine Interface (HMI) systems** that provide a centralized view of the plant's operations. They collect data from Level 1 controllers, display it to operators, and often include control functions for higher-level coordination. Operators can monitor the plant's status, make adjustments, and respond to alarms and events.
 - **Level 3 (Program Logic Controller Level):** This level is responsible for **managing and controlling** the overall production process. It often involves more advanced software systems that can coordinate multiple production lines or areas within the plant. Level 3 systems may also include functions such as recipe management, production scheduling, and data logging for analysis and reporting.

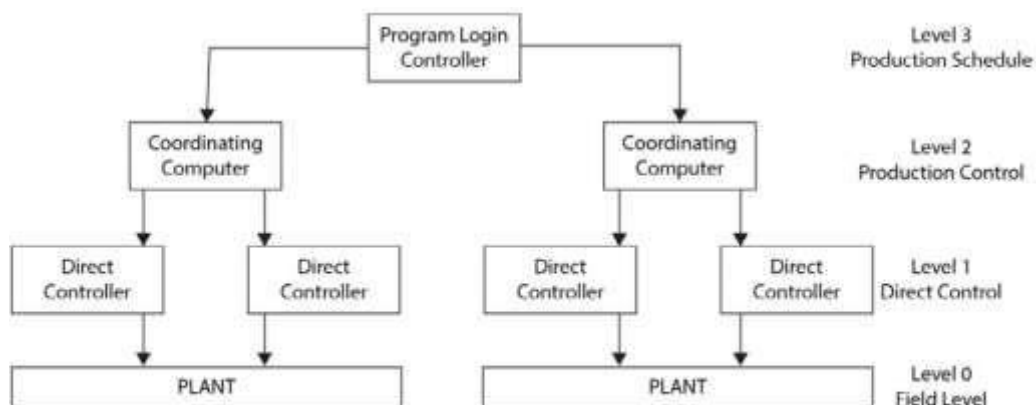


Figure 10.6: SCADA system

- SCADA systems control critical infrastructure sectors like energy, facilities (HVAC), manufacturing, logistics, and industrial processes (e.g., steel production, water treatment), making them essential for smooth operations.
 - **Energy:** Used for generating electricity and supporting oil and gas refinery operations.
 - **Facilities:** Used for building management, especially temperature control through HVAC systems.
 - **Manufacturing:** Involves producing and assembling components to create finished products like computers.
 - **Logistics:** Manages the collection and delivery of products and components from different locations to customers.
 - **Industrial:** Focuses on converting raw materials into useful outputs, such as steel production or water treatment.
- Due to their vital role, SCADA systems are high-value targets for cyberattacks, requiring robust cybersecurity measures to prevent disruptions in essential services.